

Group 1 - BIPN 162 Final Project

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Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) has been associated with disruptions in large-scale brain connectivity networks. In this project, we investigate whether a transformer-based classifier can distinguish ADHD patients from healthy controls using resting-state functional MRI (rs-fMRI) data. Functional connectivity matrices are computed from regional brain time series and used as inputs to a transformer architecture that models interactions between brain regions through multi-head self-attention. We replicate and adapt the transformer-encoder framework proposed by Dai et al. (2023) and apply it to an ADHD rs-fMRI dataset to evaluate whether the model can capture connectivity patterns that differentiate ADHD from typical brain activity. Model performance is assessed using classification metrics including accuracy, ROC-AUC, and F1-score, and results are compared with baseline machine learning approaches.

Background & Prior Work

Attention-Deficit/Hyperactivity Disorder (ADHD) is a common neurodevelopmental condition characterized by inattention, impulsivity, and executive dysfunction (Kuntsi et al., 2006). Neuroimaging studies have shown that ADHD is associated with altered functional connectivity across several large-scale brain networks, including the default mode network, frontoparietal control network, and attention networks (Silk et al., 2005). These network-level differences suggest that ADHD may involve disruptions in communication between distributed brain regions rather than localized abnormalities. Resting-state functional MRI (rs-fMRI) provides a noninvasive method for measuring spontaneous brain activity and has been widely used to investigate connectivity patterns associated with neurological and psychiatric conditions. By examining correlations between time series recorded from different brain regions, researchers can estimate functional connectivity and characterize the brain as a network of interacting regions. However, identifying reliable biomarkers from rs-fMRI remains challenging due to noise, inter-subject variability, and the high dimensionality of neuroimaging data.

Recent advances in deep learning have introduced new approaches for modeling complex relationships in high-dimensional datasets. Transformer architectures, originally developed for natural language processing, have demonstrated strong performance in tasks involving structured and relational data. Because transformers use self-attention mechanisms to model relationships between elements in a sequence, they are well suited for functional connectivity analysis where interactions between distant brain regions are important. In this project, we investigate whether a transformer-based classifier can distinguish ADHD patients from healthy controls using rs-fMRI connectivity patterns. Machine learning methods have increasingly been applied to resting-state fMRI data in an effort to identify biomarkers for ADHD and other neuropsychiatric conditions. Early approaches commonly used traditional classifiers such as support vector machines, logistic regression, and multilayer perceptrons trained on functional connectivity features. For example, the ADHD-200 Global Competition demonstrated that machine learning models could achieve moderate classification performance using neuroimaging features, although performance was often limited by dataset variability and feature engineering challenges (Brown et al., 2012).

More recently, deep learning approaches have been introduced to better capture the complex structure of brain connectivity data. Convolutional neural networks and graph-based models have been applied to functional connectivity matrices to learn representations of brain networks directly from neuroimaging data. These models can capture nonlinear relationships between brain regions and have shown improved performance compared to traditional machine learning methods. Transformer architectures have emerged as a promising approach for analyzing neuroimaging connectivity patterns. Transformers rely on a self-attention mechanism that allows the model to learn relationships between all elements in a sequence simultaneously. The paper used the encoder part of the Transformer model to extract the input data features and discarded the decoder part to classify Major Depressive Disorder. Time series data from rs-fMRI was extracted using brain atlases and Pearson correlation was computed for the Feature extraction matrix. Each feature is projected into d dimensions into a feature space, encoded by position using deterministic sine encoding, and fed into the encoder of the transformer to obtain final representation matrix Z . The classifier is composed of linear layers and applied to obtain final classification results. We will be replicating

the same Transformer-encoder model on our fMRI dataset and test its accuracy in classifying ADHD patients. We downloaded fMRI scan files of healthy individuals (130) and ADHD patients (42) from the UCLA consortium of fMRI data. Participants age 21-50 were recruited from the Los Angeles area by community advertisements and completed extensive neuropsychological testing in addition to fMRI scanning. We extracted each subject's T1 images and resting state fMRI scans. We evaluate whether attention-based models can effectively capture connectivity patterns that differentiate ADHD patients from healthy controls.

Data Preprocessing

The ds000030 fMRI dataset was downloaded from the UCLA consortium (<https://openfmri.org/dataset/ds000030/>). The sample contains 130 controls and 42 ADHD subjects. Each subject contained 4 folders(func, anat, beh, dwi). We extracted the task-rest_bold.nii.gz files from func folder of each subject and T1w.nii.gz files from the anat folder, organizing, renaming and expanding them to prepare for its preprocessing in the DPARSF Matlab toolbox used in the study. 8 control subjects and 2 ADHD subjects are removed from the dataset due to missing T1 images and resting state fMRI scans. The first 10 volumes of the BOLD signal were discarded, slice-timing correction was used to ensure the temporal consistency of images across all layers within a single scanning session. T1-weighted images were co-registered to the mean functional image using a 6 degrees-of-freedom linear transformation and then segmented into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF). At last, the signals were transformed to MNI space and the time series were filtered through a temporal bandpass filter (0.01–0.1Hz). The output FunimgARWDCF(aligned, reoriented, Dartel normalized, filtered) files from each subject were organized and prepared for feature extraction.

For each subject, brain regions were partitioned using the AAL-116 brain atlas, yielding time series signals for 116 brain regions, respectively. and the average time series for each brain region was calculated. Then, z-score normalization was applied to each time series, which may help alleviate the impact of site differences on the experiment. The features used in this study were FCs between brain regions calculated by Nilearn (<https://nilearn.github.io/stable/index.html>) using Pearson correlation. The FC matrix obtained using the AAL-116 atlas had a dimension of 116×116 .

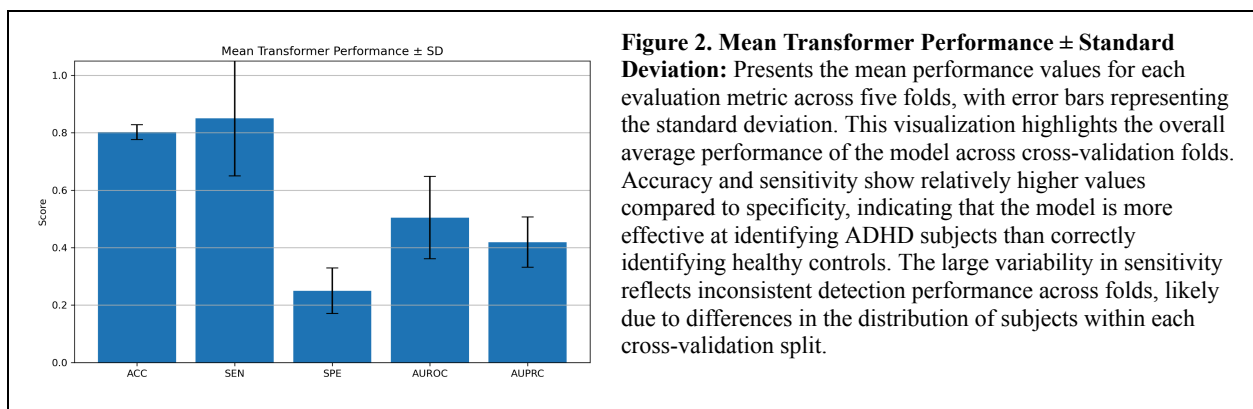
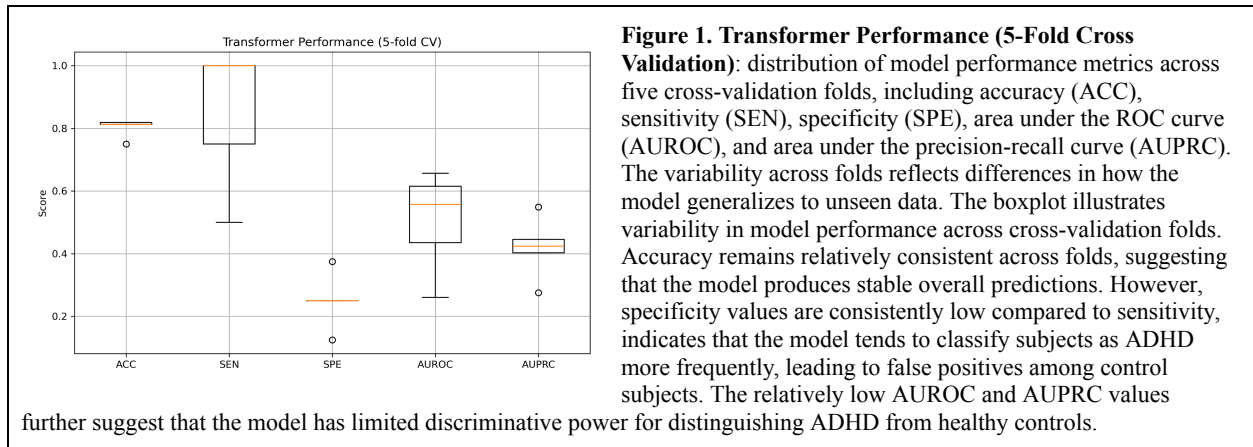
Transformer Model

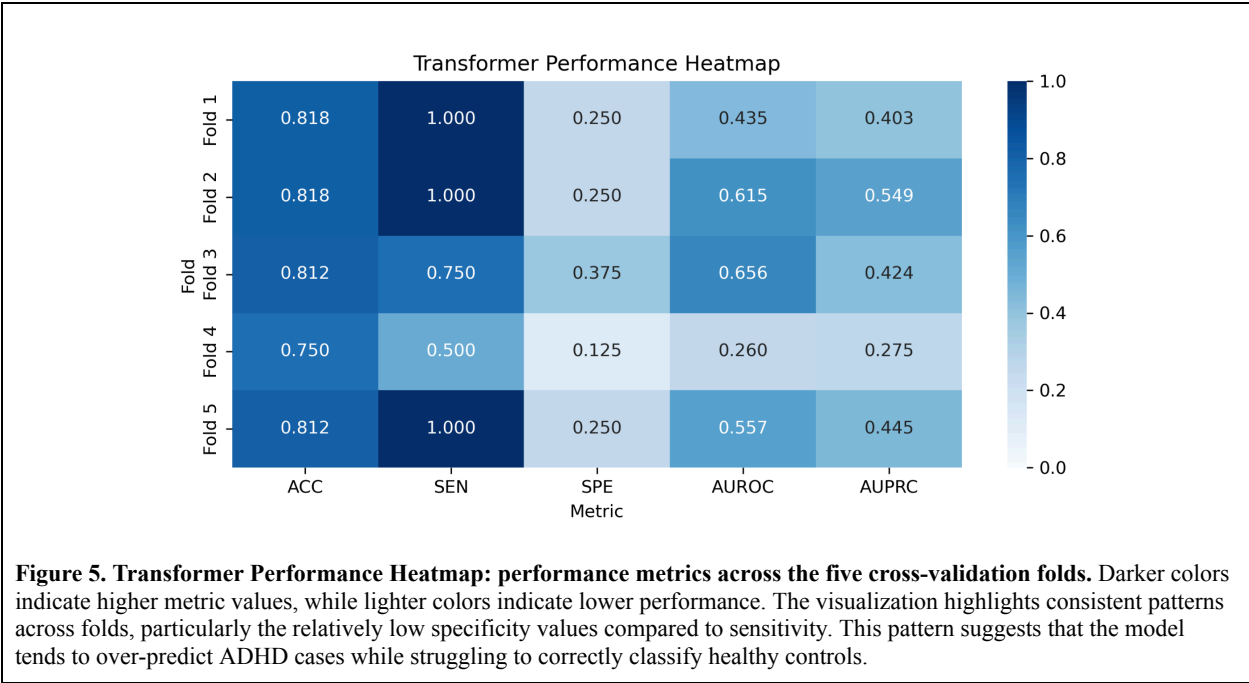
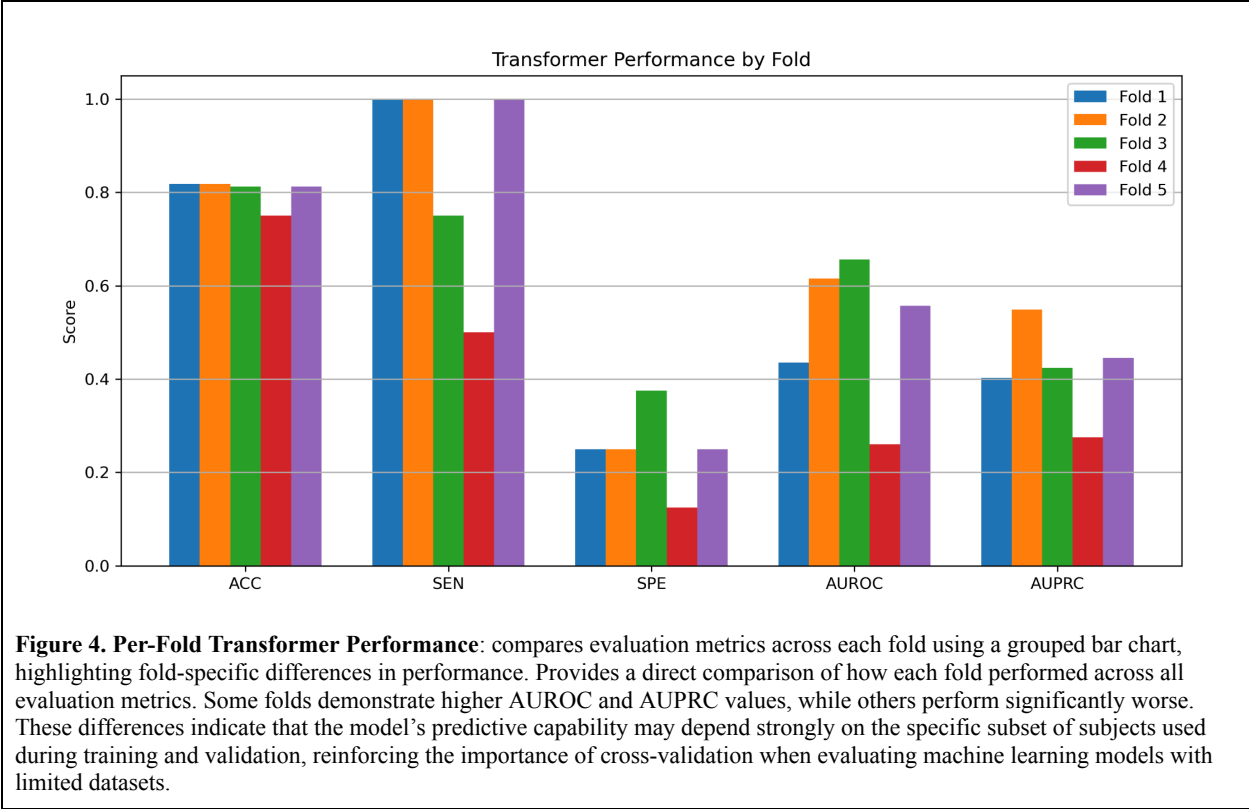
The model used in this study is based on a Transformer encoder architecture, which relies on a self-attention mechanism to model relationships between input features. Rather than using a single attention operation, the transformer applies multi-head self-attention, allowing the model to capture multiple types of relationships between brain regions simultaneously. The outputs of the attention layer are then processed by a feed-forward neural network (FNN) that applies non-linear transformations through activation functions. To improve training stability in deeper architectures, both the multi-head attention module and the feed-forward network incorporate residual connections and layer normalization, which help maintain gradient flow and prevent degradation during training.

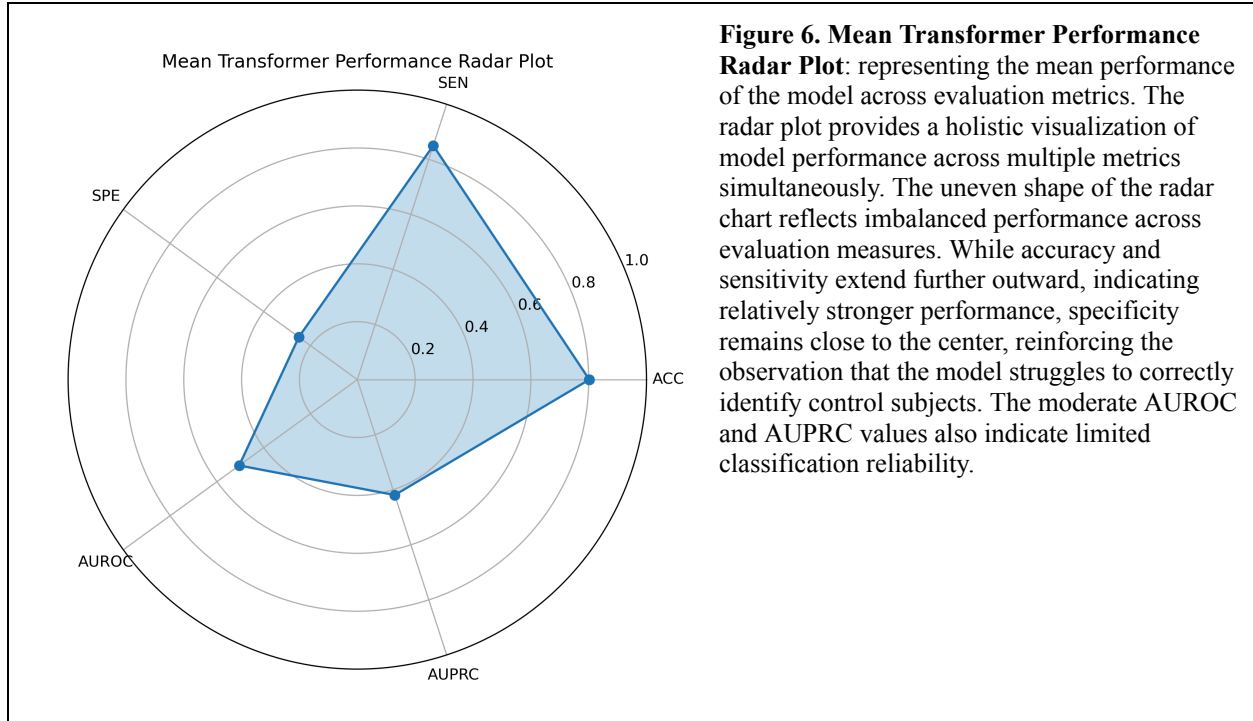
For implementation, we used PyTorch to construct and train the model. Some hyperparameters were selected through exploratory experiments, while others were informed by values reported in related studies (Deng et al., 2022; Kong et al., 2021). The network uses LeakyReLU and GELU activation functions within its layers. Model optimization was performed using the Adam optimizer with a learning rate of 0.001. To evaluate the model performance, 5-fold cross validation was used. A total of 162 datasets were randomly distributed into five folds, while each had approximately 24 controls and 8 ADHD data. Four folds were used for training and one fold was used for validation. The iteration was repeated five times so that each fold served as validation once. Training was conducted with a batch size of 32, and the model was trained for 400 epochs.

Data Visualization

To better understand the performance and behavior of the classification model, several visualizations were created to analyze cross-validation results and evaluate the model's ability to distinguish ADHD patients from healthy controls. These visualizations summarize model performance across folds and highlight variability across different evaluation metrics.







Data Analysis and Results

After training, the accuracy (ACC), sensitivity (SEN), specificity (SPE), area under the receiver operating characteristic curve (AUROC), and area under the precision-recall curve (AUPRC) were computed for each fold. ACC, SEN, and SPE were used to assess the model performance, while AUROC and AUPRC reflected the discriminative ability of the model. For analysis, the best validation performance within each fold evaluated by 5-fold cross validation was extracted. Then, the average performance across five folds was calculated.

Metric	ACC (%)	SEN (%)	SPE (%)	AUROC	AUPRC
Mean ± SD	80.2 ± 2.8	85.0 ± 22.4	25.0 ± 7.9	0.505 ± 0.146	0.419 ± 0.089

Table 1. Performance summary across five folds: ACC (%), SEN (%), SPE (%), AUROC, and AUPRC

Across five folds, the model showed high mean accuracy (80.2 ± 2.8 %) and mean sensitivity (85.0 ± 22.4 %). However, the specificity was low, being 25.0 ± 7.9 % (Table 1). Sensitivity represents the true positive rate while specificity represents the true negative rate. Together, the high sensitivity and low specificity values indicated that the model detects ADHD frequently and correctly, while the controls were often misclassified as ADHD, giving false positives.

Performance summary also revealed the limited discriminative ability. The mean AUROC score was 0.505 ± 0.146 and the AUPRC score was 0.419 ± 0.089 . The AUROC score below 0.6 indicated that the distinction between ADHD and controls were likely due to random chance. The AUPRC score well represented the precision-recall performance of the model, especially for the imbalance dataset of the experiment. The baseline AUPRC was 0.247, representing the prevalence of the positive class. While the AUPRC score did exceed the baseline, the score was still relatively moderate, implying that the classifier has limited precision-recall performance. Overall, regardless of the relatively high accuracy level, the low specificity and low AUROC score suggested that the classifier was unreliable in distinguishing ADHD from control subjects, often giving false positive results, under the use of a current number of functional connectivity features.

Conclusion

The goal of this study was to examine if a transformer-based classifier distinguished ADHD patients from healthy controls using resting-state functional MRI (rs-fMRI) by learning differences in brain functional

connectivity patterns. In order to accomplish this, the study replicated a transformer-based classification model that detects MDD using resting-state fMRI and applied it to ADHD detection. The model's performance was evaluated using 5-fold cross validation. However, it reflected the poor discriminative ability, regardless of high mean accuracy.

This suggests that the transformer did not fit well with ADHD datasets, or with disorders with more heterogeneous neural structure in general. Overall, the results highlight the complexity of machine learning models when applying to different types of disorders—that the classification models are highly dependent on dataset size and disorder-specific neural patterns.

Discussion

One possible explanation is that ADHD-specific neural patterns may be more heterogeneous than those of major depressive disorder. ADHD is highly related to the attention distribution and executive control networks, which highly vary across individuals. Hence the extracted pattern may be inconsistent, being difficult for the model to capture the pattern. Moreover, As a result, functional connectivity differences may not form a consistent pattern that can be easily captured by the model. In addition, the small sample size might limit the ability of the classifier model to precisely generalize the pattern.

There were few significant limitations in this study. First, the sample size was extremely small to train and validate a transformer. Training of machine learning models, especially the brain feature classification, typically requires large data size. Insufficient data might have decreased the effectiveness of learning complex patterns in FC matrices. Second, the data distribution was imbalanced, controls taking about 75% of the training data. This imbalance, as well as the lack of ADHD subjects, might have contributed to the bias toward ADHD predictions under minor differences. Third, only a single type of feature representation was used. Use of different feature representations might have worked better with ADHD data or provided additional information that could improve classification performance. Fourth, only a single type of brain atlas was used for feature extraction, while the original study used two different types of brain atlas with different dimensions. If the higher resolution atlas was also used, the model might have captured more neural interactions and connectivity patterns relevant to ADHD.

There are several suggestions to improve the results in the future replication. One is that the training dataset size can be increased by using more ADHD rs-fMRI datasets. This will improve the model's generalization in connectivity patterns. Second, using brain atlas with higher resolution or different references might capture better network connectivity in ADHD. It may also be useful to investigate different features other than static correlation matrices, such as dynamic functional connectivity or time resolved features. Finally, the addition of or mixing with other machine learning approaches might capture better ADHD patterns. The support vector machine (SVM) model was tested in comparison with the transformer model using the same data. While every other value was lower than the current model, the specificity was higher in SVM, suggesting alteration in the modeling pipeline.

Reflection

From the proposal to the final report, this project evolved significantly as we clarified our research goal and aligned it more closely with a specific reproducible method. Early feedback on our proposal indicated that our scientific question and background were framed too generally around machine learning classification rather than explicitly replicating a particular transformer-based approach. In response, we revised the project to clearly anchor our work to the transformer-encoder framework proposed by Dai et al. (2023) and focused on reproducing key aspects of their analysis pipeline while applying it to ADHD resting-state fMRI data.

One of the most challenging aspects of the project was understanding and implementing the preprocessing pipeline required for functional connectivity analysis. Working with high-dimensional neuroimaging data and organizing the dataset for analysis required careful attention to detail. At the same time, this process was also one of the most rewarding parts of the project because it helped us understand how neuroimaging data are transformed into features that machine learning models can use.

The other challenging part of the project was adapting the MDD transformer model to the ADHD dataset. There were nine python modules to be modified in order to run one main transformer model and 5-fold cross validation process. Adapting the file format as well as the code itself to fit the transformer while keeping the original study's method was definitely challenging for the intermediate level python user. Also, the debugging was

unexpectedly time consuming, which led to delay in progress. However, in the end, this process was very rewarding and allowed us to learn a lot about the neural data science itself as well as the transformer and classifier models.

Although some implementation details required adjustment along the way, the overall project followed our intended direction and provided valuable experience applying deep learning methods to neuroscience data.

Contributions & AI Usage Statement

Shamei Lu downloaded, organized, and preprocessed the dataset through DPARSF. Wrote up the Data preprocessing and method introduction section of the report. Feature extracted using aal atlas. Julianne Lee extracted the features and organized into FC matrix. Adapted the transformer model from the research, trained the transformer model. Validated the classifier and analyzed the performance of the model. Wrote up the data analysis and results, conclusion, and discussion sections of the report. Keerat Kaur conducted background research, wrote the abstract, introduction, prior work section of the report, created data visualizations and interpretations. AI tools (specifically ChatGPT) were used in a limited capacity to assist with editing, paraphrasing, and improving the clarity of written sections of this report. The AI was primarily used to help rephrase text, improve grammar, and organize explanations of methods such as the transformer architecture and evaluation procedures. All analysis decisions, code implementation, data preprocessing, and interpretation of results were performed by the authors. The authors reviewed and edited any AI-assisted text to ensure accuracy and alignment with the project's methods and findings.

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